Weighted Adaptive Stimulation After Batch-testing Identification (WASABI) Problem Formulation

Problem Formulation

There are a few ways we can formulate the question

1. Similar to Draelos 2020 first identify the binary connections using group testing, then use the identified connections to fit a weighted model.
2. Directly use a linear-Poisson model to identify weights using group testing. (Works on the assumption that connections are linear)
3. If not linear, use a neural network or some other nonlinear approach.

We will focus on (2) first, with fixed continuous stimulation parameters (e.g., current, frequency...)

Goal

To develop a method for adaptive stimulation that iteratively identifies the most informative stimulation sites. The dynamics of the reach is modeled using a linear-Poisson model , using a weighted linear-Poisson model to identify the most informative stimulation sites and improve the accuracy of the model.

Dynamics model

We model stimulation as a perturbation to a latent dynamical system through the mapping $\boldsymbol{B}$. Specifically, we model the system using a Poisson linear dynamical system (PLDS), which combines firing rate estimation with dimensionality reduction. We assume linear, Markovian (AR(1)) state evolution, Poisson emissions, and stationary $\boldsymbol{A}$. Eventually, we can have a Kalman-like update to allow for nonstationarity in $\boldsymbol{C}$ and a compositional model for $\boldsymbol{A}$.

State space equation

$$\begin{aligned} \boldsymbol{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), & \boldsymbol{u} &= \text{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) \\ \boldsymbol{x}_{t+1} &= \boldsymbol{A}\boldsymbol{x}_t + \boldsymbol{B}\boldsymbol{u}_t + \boldsymbol{\epsilon}_t, & \boldsymbol{\epsilon}_t &\sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\ \boldsymbol{y}_t &\sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big) \end{aligned}$$

Variables

- $\boldsymbol{x}_t \in \mathbb{R}^K$: latent state at time t
  - K is the latent dimensionality
- $\boldsymbol{y}_t \in \mathbb{Z}_{\geq 0}^N$: binned spike count vector at time t
  - N is the number of channels
- $\boldsymbol{s}_t \in \{0,1\}^N$: one-hot stim indicator at time t
  - zero vector on control trials
  - stimulation occurs at $t = t_{\text{stim}}$

Hyperparameters

- $K \in \mathbb{Z}_+$: latent dimensionality
  - $K \ll N$, low-rank assumption
- $w \in \mathbb{R}_+$: stim spatial spread radius
- $\boldsymbol{V} \in \mathbb{R}^{N \times N}$: Toeplitz matrix encoding spatial current spread

Fitted parameters

- $\boldsymbol{A} \in \mathbb{R}^{K \times K}$: dynamics operator
- $\boldsymbol{B} \in \mathbb{R}^{K \times N}$: stimulus-to-latent mapping
- $\boldsymbol{b} \in \mathbb{R}^K$: perturbation threshold
  - what stim strength would perturb latents
- $\boldsymbol{C} \in \mathbb{R}^{N \times K}$: observation (loading) matrix
- $\boldsymbol{d} \in \mathbb{R}^N$: per-channel mean log-firing rate
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\boldsymbol{\mu}_0 \in \mathbb{R}^K$, $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{K \times K}$: initial state mean and covariance
- $\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{b}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}\}$: full parameter set

$$V_{ij} = \frac{1}{Z(w)} \exp\left(-\frac{(i-j)^2}{2w^2}\right) \quad Z(w) = \sum_{k=-\infty}^{\infty} \exp\left(-\frac{k^2}{2w^2}\right)$$

Overview

1. **Control trials**: Primate performs control reaches to the target
   (a) Fit the Poisson linear dynamical system (PLDS) using Extended Kalman Filter EKF to the control reaches to obtain $\hat{\boldsymbol{\theta}}_{\text{ctrl}}$
2. **Stim trials**:
   (a) **Model fitting**: Between each trial, we fit the spike counts $\{\boldsymbol{y}_{i,1:T}\}$ to the model using the EKF + RTS smoothing to get $\{\hat{\boldsymbol{x}}_{i,1:T}, \boldsymbol{P}_{i,1:T}\}$. We also get updated parameters $\hat{\boldsymbol{\theta}}_{\text{i}} \leftarrow \hat{\boldsymbol{\theta}}_{\text{i}-1}$
   (b) At the first trial, we have no information about stim-to-latent mapping, so the first stimulation sites are chosen at random.
3. **Acquisition:**
   (a) **Recursive least squares**: Using RLS to track $\hat{\boldsymbol{B}}_i$ and $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$ gives us an estimate of the covariance in the stim-to-latent mapping.
   (b) **Stim selection**: Choose the next stimulation $\boldsymbol{s}_{i+1}$ using the covariance estimate of $\hat{\boldsymbol{B}}_{i+1|i}$ $(\hat{\boldsymbol{\Sigma}}_{B_{i+1|i}})$
4. Perform the next stimulation and repeat 2-3.

Long overview

1. **Control trials**: Monkey performs control reaches to the target
2. We fit the PLDS to the control reaches to obtain $\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}$
3. **Stim trials**: During each trial, perform online filtering using the fitted parameters $\hat{\boldsymbol{\theta}}_{\text{i}-1}$, to get $\{\boldsymbol{x}_{i,1:t}\}$ in real time. This gives us the state at which the stimulation happens.
   (a) At the first trial, we initialize with $\hat{\boldsymbol{\theta}}_{\text{i}-1} = \{\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{B}}_{0|0}, \hat{\boldsymbol{b}}_{0|0}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}, \hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\Sigma}}_0\}$, where $\hat{\boldsymbol{B}}_{0|0} = \boldsymbol{0}$, $\hat{\boldsymbol{\Sigma}}_{B_{0|0}} = \sigma_B^2 \boldsymbol{I}$, and $\hat{\boldsymbol{b}}_{0|0} = \boldsymbol{0}$.
   (b) This means that we have no information about the stim-to-latent mapping or the perturbation threshold and the first stimulation sites are chosen at random.
4. **Model fitting**: Between each trial, we fit the spike counts $\{\boldsymbol{y}_{i,1:T}\}$ to the model using the Extended Kalman Filter (EKF)-style integral and stochastic gradient descent (SGD). This gives us updated parameters $\hat{\boldsymbol{\theta}}_{\text{i}} \leftarrow \hat{\boldsymbol{\theta}}_{\text{i}-1}$
   (a) The previous trial would "warm-start" the convergence of the next trial, so we only need to run a few iterations of EKF and SGD between each trial to get a converging estimate of the parameters.
5. **Acquisition:**
   (a) **Recursive least squares**: Using RLS to track $\hat{\boldsymbol{B}}_i$ and $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$ gives us an estimate of the covariance in the stim-to-latent mapping. This step uses $\hat{\boldsymbol{A}}_i, \hat{\boldsymbol{Q}}_i, \hat{\boldsymbol{x}}_{i,t|T}, \boldsymbol{P}_{i,t|T}$, and $\hat{\boldsymbol{d}}_i$ estimated from steps 4.
   (b) **Stim selection**: Choose the next stimulation $\boldsymbol{s}_{i+1}$ using the covariance estimate of $\hat{\boldsymbol{B}}_{i+1|i}$ ($\hat{\boldsymbol{\Sigma}}_{B_{i+1|i}}$), which is the uncertainty in the stim-to-latent mapping after trial $i+1$ after a given stimulation.
6. Perform the next stimulation and repeat 3-5.

Model fitting

State-space equation (PLDS)

$$
\begin{aligned}
\boldsymbol{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\boldsymbol{u} &= \text{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) \\
\boldsymbol{x}_{t+1} &= \boldsymbol{A}\boldsymbol{x}_t + \boldsymbol{B}\boldsymbol{u}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
\boldsymbol{y}_t &\sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)
\end{aligned}
$$

We denote two distinct indices throughout: Ex: for $\boldsymbol{A}_{i,t}$

- $i = 1, 2, \ldots$ indexes *trials*
- $t = 1, \ldots, T$ indexes *within-trial timesteps*

Initialization (priors at first stim trial)

We first collect a calibration session with the monkey performing control reaches, which identifies the initial model parameters (and spike detection templates).

Control trial initialization ($i = 0$)

Initialize and fit PLDS on control trials:

$$
\begin{aligned}
\boldsymbol{A}_{0,\text{ctrl}} &= \boldsymbol{I}_K \\
\boldsymbol{C}_{0,\text{ctrl}} &\sim \mathcal{N}(\boldsymbol{0},\, \sigma_C^2\, \boldsymbol{I}_{N \times K}) \\
\boldsymbol{d}_{0,\text{ctrl}} &= \log \bar{\boldsymbol{y}}_{0,\text{ctrl}} \\
\boldsymbol{\mu}_{0,\text{ctrl}} &= \boldsymbol{0}_K \\
\boldsymbol{\Sigma}_{0,\text{ctrl}} &= \boldsymbol{I}_K
\end{aligned}
$$

$\Downarrow$ fit on control data

$$
\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}
$$

First stim trial initialization ($i = 1$)

Warm start with fitted control parameters:

$$
\begin{aligned}
\hat{\boldsymbol{A}}_1 &= \hat{\boldsymbol{A}}_{\text{ctrl}} \\
\hat{\boldsymbol{C}}_1 &= \hat{\boldsymbol{C}}_{\text{ctrl}} \\
\hat{\boldsymbol{d}}_1 &= \log \bar{\boldsymbol{y}}_1
\end{aligned}
$$

Stim parameters are initialized via:

$$
\begin{aligned}
\hat{\boldsymbol{B}}_{0|0} &= \boldsymbol{0}_{K \times N} \\
\hat{\boldsymbol{\Sigma}}^{(k)}_{B_{0|0}} &= \sigma_k^2\, \boldsymbol{I}_N \\
\hat{\boldsymbol{b}}_{0|0} &= \boldsymbol{0}_K
\end{aligned}
$$

where:

- σ_k^2: prior uncertainty over row k of the stim-to-latent mapping; set $\sigma_k^2 = \sigma_B^2$ for all k to recover a single shared scale
- σ_C^2: controls prior uncertainty over the latent-to-observation mapping
- $\bar{\boldsymbol{y}}_{\text{ctrl}}$: mean spike count from control trials
- a spherical prior on each row means $\boldsymbol{s}_1$ is chosen at random

Stim trial initialization ($i \geq 2$) uses the previous trial's fitted parameters $\hat{\boldsymbol{\theta}}_{i-1}$ as the prior for the next trial.

Online forward filtering during a trial

During a trial, we estimate the latent state $\boldsymbol{x}_{i,t}$ in real time using the EKF forward pass with the current parameter estimates $\hat{\boldsymbol{\theta}}_{i-1}$. This gives us an estimate of the latent state in real time.

EKF forward pass

The marginal likelihood of the latent trajectory is:

$$\log p(\boldsymbol{y}_{1:T}|\boldsymbol{\theta}) = \log \int p(\boldsymbol{y}_{1:T} \mid \boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, p(\boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, d\boldsymbol{x}_{1:T}$$

$$\boldsymbol{y}_t \sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)$$

$$\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{b}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}\}$$

Because the Poisson emission $p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}; \boldsymbol{\theta})$ is non-Gaussian, the integral is intractable. At time t, we need to approximate it by linearizing the emission around $\hat{\boldsymbol{x}}_{t+1|t}$, the current filtered mean (latent state), yielding a Gaussian approximation to the predictive distribution

$$p(\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t}; \boldsymbol{\theta}) \approx \mathcal{N}(\boldsymbol{\lambda}_{t+1|t},\, \boldsymbol{S}_t), \qquad \boldsymbol{\lambda}_{t+1|t} = \text{softplus}\Big(\boldsymbol{C}\hat{\boldsymbol{x}}_{t+1|t} + \boldsymbol{d}\Big)$$

where $\boldsymbol{S}_t$ is the estimated covariance.

$$\mathcal{L}(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_t|\, \boldsymbol{\lambda}_{t+1|t},\, \boldsymbol{S}_t)$$

Emission Jacobian

In the EKF formulation, we approximate the observation operator $\boldsymbol{H}_t$ with the Emission Jacobian described here: For $\boldsymbol{\lambda}_t = \text{softplus}(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d})$, we have the Jacobian (log-rate with respect to the latent state) is:

$$\boldsymbol{H}_t = \left.\frac{\partial \boldsymbol{\lambda}_t}{\partial \boldsymbol{x}_t}\right|_{\hat{\boldsymbol{x}}_t} = \text{diag}(\sigma(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d}))\, \boldsymbol{C} \in \mathbb{R}^{N \times K}$$

where $\sigma(\cdot)$ is the sigmoid function (derivative of softplus). We calculate the Emission Jacobian $\boldsymbol{H}_t$ to linearize the non-Gaussian Poisson emission and plug into the Kalman equations. This propagates the Gaussian estimations of latent states $\boldsymbol{x}_t$ one time step forward each, which we can plug in the Kalman equations.

EKF forward pass prediction update

$$\begin{aligned}
\hat{\boldsymbol{x}}_{t|t-1} &= \boldsymbol{A}\hat{\boldsymbol{x}}_{t-1|t-1} + \boldsymbol{B}\,\text{softplus}(\boldsymbol{V}\boldsymbol{s}_{t-1} + \boldsymbol{b}) \\
\boldsymbol{P}_{t|t-1} &= \boldsymbol{A}\boldsymbol{P}_{t-1|t-1}\boldsymbol{A}^\top + \boldsymbol{Q} \\
\boldsymbol{S}_t &= \boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top + \text{diag}(\boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{K}_t &= \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1} \\
\hat{\boldsymbol{x}}_{t|t} &= \hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{P}_{t|t} &= (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H}_t)\boldsymbol{P}_{t|t-1}
\end{aligned}$$

- $\hat{\boldsymbol{x}}_{t|t-1} \in \mathbb{R}^K$: predicted latent state mean
- $\hat{\boldsymbol{x}}_{t|t} \in \mathbb{R}^K$: filtered latent state mean
- $\boldsymbol{P}_{t|t-1} \in \mathbb{R}^{K \times K}$: predicted latent state covariance
- $\boldsymbol{P}_{t|t} \in \mathbb{R}^{K \times K}$: filtered latent state covariance
- $\boldsymbol{S}_t \in \mathbb{R}^{N \times N}$: innovation covariance
- $\boldsymbol{K}_t \in \mathbb{R}^{K \times N}$: Kalman gain
- $(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1})$: innovation (residual)
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\text{diag}(\boldsymbol{\lambda}_{t|t-1})$: Poisson observation noise
- $\boldsymbol{H}_t \in \mathbb{R}^{N \times K}$: locally linear approximation of emission matrix

Offline parameter update (Outline)

At the end of each trial i, the following steps are performed in order:

1. "E step" **EKF forward pass + RTS smoother**: Run the EKF forward over $\boldsymbol{y}_{i,1:T}$ with $\hat{\boldsymbol{\theta}}_{i-1}$ fixed, then run the RTS smoother backward to get filtered means and covariances $\{\hat{\boldsymbol{x}}_{i,1:T}, \boldsymbol{P}_{i,1:T}\}$. $\{\hat{\boldsymbol{x}}_{i,t|t}, \boldsymbol{P}_{i,t|t}\}_{t=1}^{T}$ is the lag-one cross-covariance needed for the M-steps below.

2. "M step" **Factorized estimation of parameters**: We factorize the estimation of parameters to get point estimates of parameters with no closed form ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$, $\hat{\boldsymbol{b}}_i$), but still getting covariance estimates for those that we can using EM. This also keeps the faster convergence rate of the EM algorithm.

   (a) **SGD update $\hat{\boldsymbol{\theta}}_i$** Take one gradient step on the single-trial EKF marginal log-likelihood $\mathcal{L}_i = \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_{i,t} \mid \boldsymbol{\lambda}_{i,t|t-1}, \boldsymbol{S}_{i,t})$:

$$\hat{\boldsymbol{\theta}}_i \leftarrow \hat{\boldsymbol{\theta}}_{i-1} - \eta_i \nabla_{\boldsymbol{C},\boldsymbol{d}} \left[-\mathcal{L}_i(\hat{\boldsymbol{\theta}}_{i-1})\right]$$

3. **Repeat 1 and 2 until convergence**:

   (a) Steps (a)-(c) are handled in the DYNAMAX package using conditional moments gaussian smoother (CMGS), but we lay out the details below.

   (b) We may need hyperparameter optimization on the batch size and learning rate.

   (c) This process can terminate early if the next trial happens before convergence.

   (d) We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation)

EKF forward pass + RTS smoother

The EKF forward pass is identical to the online version. (see **Online forward filtering**)

RTS smoother backward pass

$$
\begin{aligned}
\boldsymbol{G}_t &= \boldsymbol{P}_{t|t}\boldsymbol{A}^\top \boldsymbol{P}_{t+1|t}^{-1} \\
\hat{\boldsymbol{x}}_{t|T} &= \hat{\boldsymbol{x}}_{t|t} + \boldsymbol{G}_t\Big(\hat{\boldsymbol{x}}_{t+1|T} - \hat{\boldsymbol{x}}_{t+1|t}\Big) \\
\boldsymbol{P}_{t|T} &= \boldsymbol{P}_{t|t} + \boldsymbol{G}_t\Big(\boldsymbol{P}_{t+1|T} - \boldsymbol{P}_{t+1|t}\Big)\,\boldsymbol{G}_t^\top
\end{aligned}
$$

initialized at $\hat{\boldsymbol{x}}_{T|T}$, $\boldsymbol{P}_{T|T}$ and run backward for $t = T-1, \ldots, 1$.

SGD parameter update $\hat{\boldsymbol{\theta}}_i$

Parameters related to the Poisson observation equation ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$) and the stim-to-latent threshold ($\hat{\boldsymbol{b}}_i$) have no closed form. So, we made a design choice to update the parameter set by using SGD and autodiff through the posterior:

$$
\boldsymbol{\theta}^{(\tau+1)} = \boldsymbol{\theta}^{(\tau)} - \eta_\tau \, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right]
$$

where η_τ is the effective learning rate defined by the Adam optimizer. More specifically:

$$
\begin{aligned}
\boldsymbol{m}_\tau &= \beta_1 \boldsymbol{m}_{\tau-1} + (1-\beta_1)\,\nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big] && \text{(momentum)} \\
\boldsymbol{v}_\tau &= \beta_2 \boldsymbol{v}_{\tau-1} + (1-\beta_2)\,\nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big]^2 && \text{(RMSprop)} \\
\hat{\boldsymbol{m}}_\tau &= \frac{\boldsymbol{m}_\tau}{1-\beta_1^\tau}, \qquad \hat{\boldsymbol{v}}_\tau = \frac{\boldsymbol{v}_\tau}{1-\beta_2^\tau} && \text{(bias correction)} \\
\boldsymbol{\theta}^{(\tau+1)} &= \boldsymbol{\theta}^{(\tau)} - \frac{\alpha}{\sqrt{\hat{\boldsymbol{v}}_\tau} + \epsilon}\,\hat{\boldsymbol{m}}_\tau && \text{(parameter update)}
\end{aligned}
$$

where $\boldsymbol{m}_\tau$ and $\boldsymbol{v}_\tau$ are the first and second moment estimates, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ are the default hyperparameters, and α is the base learning rate. Gradients are computed via automatic differentiation in JAX in the DYNAMAX package. We get $\hat{\boldsymbol{\theta}}_i$ after convergence.

Getting the posterior $p(\hat{\boldsymbol{B}}_{i|i-1} \mid \boldsymbol{y}_{1:i-1}, \boldsymbol{s}_i)$

Stimulation pattern $\boldsymbol{s}_i \in \{0,1\}^N$ is fixed for trial i and happens at t_{stim}.

$$\begin{aligned}
\boldsymbol{x}_{i,0} &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\boldsymbol{x}_{i,t} &= \boldsymbol{A}\boldsymbol{x}_{i,t-1} + \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
\boldsymbol{y}_{i,t} &\sim \text{Poisson}\Big(\text{softplus}\big(\boldsymbol{C}\boldsymbol{x}_{i,t} + \boldsymbol{d}\big)\Big)
\end{aligned}$$

We assume we can factorize across the $\boldsymbol{\theta}$s (Much like coordinate ascent variational inference). We get the estimates of $\boldsymbol{\theta} \setminus \boldsymbol{B}$ known from PLDS fit via EKF SGD from trials $1, \ldots, i-1$ and treated as fixed at trial i. $\boldsymbol{s}_t$ is known:

$$\boldsymbol{\theta} \setminus \boldsymbol{B} = \{\boldsymbol{A},\, \boldsymbol{C},\, \boldsymbol{d},\, \boldsymbol{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\} = \hat{\boldsymbol{\theta}}_{i-1} \setminus \boldsymbol{B}$$

$$\begin{aligned}
p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t}) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) \\
p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{B},\, \boldsymbol{\theta} \setminus \boldsymbol{B}) \cdot p(\boldsymbol{B}) \cdot p(\boldsymbol{A}) \cdot p(\boldsymbol{C}) \cdots \quad \text{(by independence of } \boldsymbol{\theta} \text{ components)}
\end{aligned}$$

Absorbing $p(\boldsymbol{\theta} \setminus \boldsymbol{B})$ into the constant and expanding over $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) \propto \underbrace{p(\boldsymbol{B})}_{\text{Initialize via } \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_0^B)} \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{B}, \boldsymbol{\theta} \setminus \boldsymbol{B})}_{\text{Poisson emission}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{depends on } \boldsymbol{B}, \text{can get from EKF}} d\boldsymbol{x}_{1:t} \tag{1}$$

Likelihood after marginalizing out $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) = \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{\theta})}_{\text{Poisson emission, need EKF approx.}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{Gaussian dynamics prior}} d\boldsymbol{x}_{1:t}$$

$$\boldsymbol{\theta} = \{\boldsymbol{A},\, \boldsymbol{B},\, \boldsymbol{C},\, \boldsymbol{d},\, \boldsymbol{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\}$$

Bayes':

$$\underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t})}_{\text{posterior}} = \frac{\overbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\boldsymbol{y}_{1:t})}_{\text{evidence, constant in } \boldsymbol{\theta}}}$$

Acquisition

We select the next stimulation pattern by minimizing either the determinant (D-optimal) or the trace (A-optimal) of the covariance of $\boldsymbol{B}$. From SGD, we only get point estimates of the PLDS parameters. Therefore, we run a separate recursive least squares (RLS) tracker for $\boldsymbol{B}$. We work with two representations (assumptions) of that posterior covariance:

1. **Per row covariance**: K separate $N \times N$ matrices $\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}$, one per latent, assuming stimulation effects on the latents do not interact. We use this for **D-optimal**, since the determinant calculation for the full covariance may be intractable. The per row log-determinant decomposes into a sum of K terms.

2. **Full $NK \times NK$ covariance** over $\boldsymbol{B}$ stacked into a vector. Retains cross-latent correlations. We use this for **A-optimal**, which doesn't need a determinant of a large matrix.

   (a) TODO: The trace reads only diagonal entries, but its one-trial-ahead *value* still depends on the off-diagonal blocks through the update, so the two pairings are not interchangeable.

One-step ahead prediction of latents gives us the estimate of the perturbation

Given the smoothed latents $\{\hat{\boldsymbol{x}}_{i,t|T}\}_{t=1}^{T}$, we can treat each row as a scalar observation RLS. We describe it in the following:

The effective stimulation input is $\boldsymbol{u}_i = \text{softplus}(\boldsymbol{V}\boldsymbol{s}_i + \boldsymbol{b}) \in \mathbb{R}^N$. The stimulation-induced perturbation is $\boldsymbol{z}_i = \boldsymbol{B}\boldsymbol{u}_i$, and from the EKF forward pass + RTS smoother at trial i we estimate it by the one-step residual. Here, t denotes the final time bin at which the stimulation input is applied, so the stimulation-induced perturbation is observed in the transition from t to $t+1$.

$$
\begin{aligned}
\hat{\boldsymbol{z}}_i &:= \hat{\boldsymbol{x}}_{i,t+1|T} - \hat{\boldsymbol{A}}_i \hat{\boldsymbol{x}}_{i,t|T} \\
\boldsymbol{z}_i &= \boldsymbol{B}\boldsymbol{u}_i \\
\boldsymbol{z}_i &= \hat{\boldsymbol{x}}_{i,t+1|T} - \hat{\boldsymbol{A}}_i \hat{\boldsymbol{x}}_{i,t|T} - \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_i),
\end{aligned}
$$

The error covariance collects the smoother uncertainty at both timesteps plus the process noise:

$$
\begin{aligned}
\boldsymbol{R}_i &= \text{Cov}(\hat{\boldsymbol{x}}_{i,t+1|T}) + \hat{\boldsymbol{A}}_i \text{Cov}(\hat{\boldsymbol{x}}_{i,t|T}) \hat{\boldsymbol{A}}_i^\top + \hat{\boldsymbol{Q}}_i \\
&\quad - \text{Cov}(\hat{\boldsymbol{x}}_{i,t+1|T}, \hat{\boldsymbol{x}}_{i,t|T}) \hat{\boldsymbol{A}}_i^\top - \hat{\boldsymbol{A}}_i \text{Cov}(\hat{\boldsymbol{x}}_{i,t+1|T}, \hat{\boldsymbol{x}}_{i,t|T})^\top \\
&\quad \text{(Assumption: we don't have cross-cov, drop for now, and we ignore } \text{Cov}(\hat{\boldsymbol{A}})\text{)} \\
&\approx \hat{\boldsymbol{P}}_{i,t+1|T} + \hat{\boldsymbol{A}}_i \hat{\boldsymbol{P}}_{i,t|T} \hat{\boldsymbol{A}}_i^\top + \hat{\boldsymbol{Q}}_i
\end{aligned}
$$

Notation

Symbol	Definition
$\boldsymbol{w}^{(k)} \in \mathbb{R}^N$	channel weights driving latent k; $\boldsymbol{B} = \boldsymbol{W}^\top$
$\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \in \mathbb{R}^{N \times N}$	per-row covariance of row k after trial i
$\boldsymbol{\beta} = \text{vec}(\boldsymbol{W}) \in \mathbb{R}^{NK}$	$\boldsymbol{B}$ as a single parameter vector TODO
$\boldsymbol{\Sigma}_{B_i} \in \mathbb{R}^{NK \times NK}$	full covariance; blocks $\boldsymbol{\Sigma}_{B_i}^{(k,l)} = \text{Cov}(\boldsymbol{w}^{(k)}, \boldsymbol{w}^{(l)})$, $\boldsymbol{\Sigma}_{B_i}^{(k,k)} = \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}$
σ_k^2	prior scale for row k
$\sigma_{k\|\boldsymbol{u}_i}^2 = \boldsymbol{u}_i^\top \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i$	prior variance of row k along the stimulated direction
$\boldsymbol{D}_i = \boldsymbol{M}_\tau \, \text{diag}(\sigma_{1\|\boldsymbol{u}_i}^2, \ldots, \sigma_{K\|\boldsymbol{u}_i}^2) \, \boldsymbol{M}_\tau^\top$	parameter-uncertainty part of the predictive covariance
$\boldsymbol{R}_i$	smoother noise covariance
$\boldsymbol{S}_i = \boldsymbol{D}_i + \boldsymbol{R}_i \in \mathbb{R}^{K \times K}$	predictive covariance of $\boldsymbol{z}_i$; the only matrix inverted
$\boldsymbol{M}_\tau$	accumulated dynamics operator over the stimulation duration
$\boldsymbol{\eta}_i = \hat{\boldsymbol{z}}_i - \boldsymbol{M}_\tau \hat{\boldsymbol{B}}_{i-1} \boldsymbol{u}_i$	innovation at trial i

Multiple step ahead perturbation gives us perturbations propagated by the dynamics operator

If stimulation happens over multiple time bins where we have missing observations, we treat the prestim and poststim observation as a multi-step propagation. The stimulation-induced perturbation is propagated through the dynamics operator over the stimulation duration. Let τ denote the number of stimulation bins. Then

$$\boldsymbol{M}_\tau := \boldsymbol{I} + \boldsymbol{A} + \cdots + \boldsymbol{A}^{\tau-1}.$$

The effective perturbation accumulated over the stimulation period is therefore

$$\boldsymbol{z}_i = \boldsymbol{M}_\tau \boldsymbol{B} \boldsymbol{u}_i.$$

If $\hat{\boldsymbol{x}}_{i,t|T}$ denotes the smoothed latent immediately before the stimulation period and $\hat{\boldsymbol{x}}_{i,t+\tau|T}$ denotes the smoothed latent immediately after stimulation, the one-step residual is replaced by the multi-step residual

$$\hat{\boldsymbol{z}}_i := \hat{\boldsymbol{x}}_{i,t+\tau|T} - \hat{\boldsymbol{A}}_i^\tau \hat{\boldsymbol{x}}_{i,t|T}$$

Thus,

$$\begin{aligned}
&\boldsymbol{z}_i = \boldsymbol{M}_\tau \boldsymbol{B} \boldsymbol{u}_i, \\
&\boldsymbol{z}_i = \hat{\boldsymbol{x}}_{i,t+\tau|T} - \hat{\boldsymbol{A}}_i^\tau \hat{\boldsymbol{x}}_{i,t|T} - \boldsymbol{\varepsilon}_i, \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_i).
\end{aligned}$$

The error covariance collects the smoother uncertainty at both endpoints plus the process noise accumulated over the stimulation period:

$$\begin{aligned}
\boldsymbol{R}_i = \operatorname{Cov}(\hat{\boldsymbol{x}}_{i,t+\tau|T}) &+ \hat{\boldsymbol{A}}_i^\tau \operatorname{Cov}(\hat{\boldsymbol{x}}_{i,t|T}) \left(\hat{\boldsymbol{A}}_i^\tau\right)^\top \\
&+ \sum_{j=0}^{\tau-1} \hat{\boldsymbol{A}}_i^j \hat{\boldsymbol{Q}}_i \left(\hat{\boldsymbol{A}}_i^j\right)^\top \\
&- \text{cross-covariance terms}
\end{aligned}$$

Under the assumption that we do not have the required cross-covariances, and ignoring $\operatorname{Cov}(\hat{\boldsymbol{A}}_i)$,

$$\begin{aligned}
\boldsymbol{R}_i \approx \hat{\boldsymbol{P}}_{i,t+\tau|T} &+ \hat{\boldsymbol{A}}_i^\tau \hat{\boldsymbol{P}}_{i,t|T} \left(\hat{\boldsymbol{A}}_i^\tau\right)^\top \\
&+ \sum_{j=0}^{\tau-1} \hat{\boldsymbol{A}}_i^j \hat{\boldsymbol{Q}}_i \left(\hat{\boldsymbol{A}}_i^j\right)^\top
\end{aligned}$$

D-optimal (per row)

D-optimal experimental design minimizes the determinant of the covariance of $\boldsymbol{B}$ (the volume of the uncertainty ellipsoid). Here we take the per-row representation: the rows of the posterior $\boldsymbol{B}$ are assumed to be row-independent, with covariance $\boldsymbol{\Sigma}_{B_i} = \mathrm{diag}(\boldsymbol{\Sigma}_{B_i}^{(1)}, \boldsymbol{\Sigma}_{B_i}^{(2)}, \ldots, \boldsymbol{\Sigma}_{B_i}^{(k)})$, so each latent carries its own $N \times N$ covariance and the log-determinant decomposes into a sum of K terms.

Row decomposition and priors

Let $\boldsymbol{w}^{(k)} \in \mathbb{R}^N$ be:

$$\boldsymbol{B} = \boldsymbol{W}^\top \in \mathbb{R}^{K \times N} = \begin{bmatrix} \boldsymbol{w}_1^\top \\ \boldsymbol{w}_2^\top \\ \vdots \\ \boldsymbol{w}_K^\top \end{bmatrix}.$$

Row k collects the effect of stimulation at each of the N channels on latent k alone. Each row has its own prior,

$$\boldsymbol{w}^{(k)} \sim \mathcal{N}\big(\mathbf{0},\, \hat{\boldsymbol{\Sigma}}_{B_0}^{(k)}\big), \qquad \hat{\boldsymbol{\Sigma}}_{B_0}^{(k)} = \sigma_k^2\, \boldsymbol{I}_N,$$

isotropic across channels, σ_k^2 is the prior "scale" for how strongly stimulation is expected to move latent k. We initialize this to be proportional to the "magnitude" of row k of $\boldsymbol{B}$. Setting all $\sigma_{B,k}^2 = \sigma_B^2$ recovers a shared variance scale. Treating the rows as independent is the assumption that stimulation effects on each latent do not interact. The only coupling that remains enters through the shared noise covariance $\boldsymbol{R}_i$.

Predictive covariance of the observation

For row k, the relevant predictive quantity is the variance of the scalar $\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i$ under the current prior $\boldsymbol{w}^{(k)} \sim \mathcal{N}(\hat{\boldsymbol{w}}_{i-1}^{(k)}, \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)})$. The variance of this predictive quantity is:

$$\begin{aligned} \mathrm{Var}(\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i) &= \mathbb{E}\Big[(\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i)^2\Big] - \big(\mathbb{E}[\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i]\big)^2 \\ &= \mathbb{E}\Big[\boldsymbol{u}_i^\top \boldsymbol{w}^{(k)}\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i\Big] - \boldsymbol{u}_i^\top\, \mathbb{E}[\boldsymbol{w}^{(k)}]\mathbb{E}[\boldsymbol{w}^{(k)}]^\top\, \boldsymbol{u}_i \\ &= \boldsymbol{u}_i^\top \Big(\mathbb{E}\Big[\boldsymbol{w}^{(k)}\boldsymbol{w}^{(k)\top}\Big] - \mathbb{E}[\boldsymbol{w}^{(k)}]\mathbb{E}[\boldsymbol{w}^{(k)}]^\top\Big)\boldsymbol{u}_i = \boldsymbol{u}_i^\top\, \mathrm{Cov}(\boldsymbol{w}^{(k)})\, \boldsymbol{u}_i. \end{aligned}$$

Using $\mathrm{Cov}(\boldsymbol{w}^{(k)}) = \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}$ from the prior, this gives

$$\sigma_{k|\boldsymbol{u}_i}^2 := \mathrm{Var}(\boldsymbol{w}^{(k)\top}\boldsymbol{u}_i) = \boldsymbol{u}_i^\top \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i,$$

Without multi-step propagation, these row-wise uncertainties enter different coordinates of $\boldsymbol{z}_i$ independently. With multi-step propagation, however, the dynamics operator mixes these latent coordinates. Specifically,

$$\boldsymbol{z}_i = \boldsymbol{M}_\tau \boldsymbol{B} \boldsymbol{u}_i.$$

Therefore the parameter-uncertainty contribution to the predictive covariance is

$$\boldsymbol{D}_i := \boldsymbol{M}_\tau\, \mathrm{diag}\left(\sigma_{1|\boldsymbol{u}_i}^2, \ldots, \sigma_{K|\boldsymbol{u}_i}^2\right) \boldsymbol{M}_\tau^\top.$$

Adding the observation noise gives the full predictive covariance

$$\begin{aligned}\mathbb{E}[\boldsymbol{z}_i \mid \boldsymbol{u}_i] &= \boldsymbol{M}_\tau \boldsymbol{B} \boldsymbol{u}_i, \\ \mathrm{Var}(\boldsymbol{z}_i \mid \boldsymbol{u}_i) &= \boldsymbol{S}_i := \boldsymbol{D}_i + \boldsymbol{R}_i \in \mathbb{R}^{K \times K}.\end{aligned}$$

The two components are:

1. $\boldsymbol{D}_i$: uncertainty in the prediction of $\boldsymbol{z}_i$ due to uncertainty in $\boldsymbol{B}$. The covariance is diagonal before propagation, but $\boldsymbol{M}_\tau$ can introduce correlations across latent coordinates.
2. $\boldsymbol{R}_i$: uncertainty in the observation of $\boldsymbol{z}_i$ from noise in the measurement process, inherited from the latents. This noise can correlate across latents.

RLS gain and update via joint Gaussian conditioning

The unknown row $\boldsymbol{w}^{(k)}$ is Gaussian and $\boldsymbol{z}_i = \boldsymbol{M}_\tau \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\varepsilon}_i$ is a linear function of it plus independent Gaussian noise, so $(\boldsymbol{w}^{(k)}, \boldsymbol{z}_i)$ are jointly Gaussian and the posterior for row k is available in closed form:

$$\boldsymbol{w}^{(k)} \mid \boldsymbol{z}_i \;\sim\; \mathcal{N}\Big(\hat{\boldsymbol{w}}_{i-1}^{(k)} + \boldsymbol{\Gamma}_i^{(k)} \boldsymbol{S}_i^{-1}\big(\boldsymbol{z}_i - \boldsymbol{M}_\tau \hat{\boldsymbol{B}}_{i-1}\boldsymbol{u}_i\big),\;\; \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} - \boldsymbol{\Gamma}_i^{(k)} \boldsymbol{S}_i^{-1} \boldsymbol{\Gamma}_i^{(k)\top}\Big),$$

with the prior mean $\hat{\boldsymbol{w}}_{i-1}^{(k)}$ and covariance $\hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}$, the predictive mean $\mathbb{E}[\boldsymbol{z}_i] = \boldsymbol{M}_\tau \hat{\boldsymbol{B}}_{i-1}\boldsymbol{u}_i$ and covariance $\boldsymbol{S}_i = \boldsymbol{D}_i + \boldsymbol{R}_i$ from above, and the cross-covariance $\boldsymbol{\Gamma}_i^{(k)} := \mathrm{Cov}(\boldsymbol{w}^{(k)}, \boldsymbol{z}_i) \in \mathbb{R}^{N\times K}$. The posterior mean and covariance are $\hat{\boldsymbol{w}}_i^{(k)}$ and $\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}$.

Cross-covariance. The l-th column of $\boldsymbol{\Gamma}_i^{(k)}$ is

$$\mathrm{Cov}\left(\boldsymbol{w}^{(k)}, [\boldsymbol{M}_\tau \boldsymbol{B}\boldsymbol{u}_i]_l + \varepsilon_{i,l}\right).$$

Expanding coordinate l of the propagated perturbation as a sum over latents, and using $[\boldsymbol{B}\boldsymbol{u}_i]_j = \boldsymbol{w}^{(j)\top}\boldsymbol{u}_i$,

$$[\boldsymbol{M}_\tau \boldsymbol{B}\boldsymbol{u}_i]_l = \sum_{j=1}^{K} [\boldsymbol{M}_\tau]_{lj}\, [\boldsymbol{B}\boldsymbol{u}_i]_j = \sum_{j=1}^{K} [\boldsymbol{M}_\tau]_{lj}\, \boldsymbol{w}^{(j)\top}\boldsymbol{u}_i.$$

The noise is independent of $\boldsymbol{B}$, so $\mathrm{Cov}(\boldsymbol{w}^{(k)}, \varepsilon_{i,l}) = \boldsymbol{0}$, and the rows are independent a priori, so $\mathrm{Cov}(\boldsymbol{w}^{(k)}, \boldsymbol{w}^{(j)}) = \boldsymbol{0}$ for $j \neq k$. Only the $j = k$ term survives:

$$\begin{aligned}\mathrm{Cov}\big(\boldsymbol{w}^{(k)}, z_{i,l}\big) &= \sum_{j=1}^{K} [\boldsymbol{M}_\tau]_{lj}\, \mathrm{Cov}\big(\boldsymbol{w}^{(k)},\, \boldsymbol{w}^{(j)\top}\boldsymbol{u}_i\big) \\ &= [\boldsymbol{M}_\tau]_{lk}\, \mathrm{Cov}\big(\boldsymbol{w}^{(k)}\big)\, \boldsymbol{u}_i = [\boldsymbol{M}_\tau]_{lk}\, \boldsymbol{v}_k, \qquad \boldsymbol{v}_k := \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \in \mathbb{R}^N.\end{aligned}$$

Every column is therefore the same vector $\boldsymbol{v}_k$, scaled by the corresponding entry of the k-th *column* of $\boldsymbol{M}_\tau$. Stacking $l = 1, \ldots, K$ gives $\boldsymbol{\Gamma}_i^{(k)} = \boldsymbol{v}_k\, (\boldsymbol{M}_\tau \boldsymbol{e}_k)^\top$, with dimensions $(N\times N)(N\times 1)(1\times K)(K\times K) = N\times K$.

Written out, the cross-covariance is

$$\boldsymbol{\Gamma}_i^{(k)} = \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \in \mathbb{R}^{N\times K}.$$

Innovation, mean update, and gain The innovation is the difference between the trial's observed perturbation and the predicted perturbation from the previous trial $i-1$. Define

$$\boldsymbol{\eta}_i := \hat{\boldsymbol{z}}_i - \boldsymbol{M}_\tau \hat{\boldsymbol{B}}_{i-1}\boldsymbol{u}_i \in \mathbb{R}^K$$

Substituting $\boldsymbol{\Gamma}_i^{(k)} = \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top$,

$$\hat{\boldsymbol{w}}_i^{(k)} = \hat{\boldsymbol{w}}_{i-1}^{(k)} + \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i^{-1} \boldsymbol{\eta}_i.$$

Define the projected gain

$$\boldsymbol{g}_{i,k} := \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i^{-1},$$

so that

$$\boxed{\hat{\boldsymbol{w}}_i^{(k)} = \hat{\boldsymbol{w}}_{i-1}^{(k)} + \boldsymbol{g}_{i,k} \boldsymbol{\eta}_i}.$$

Covariance update. The covariance update is

$$\begin{aligned}\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} &= \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} - \boldsymbol{\Gamma}_i^{(k)} \boldsymbol{S}_i^{-1} \boldsymbol{\Gamma}_i^{(k)\top} \\ &= \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} - \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \left(\boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k \right) \boldsymbol{u}_i^\top \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}.\end{aligned}$$

Let

$$\alpha_{i,k} := \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k.$$

$$\boxed{\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} = \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} - \alpha_{i,k} \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{u}_i^\top \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}}.$$

Repeating for every $k = 1, \ldots, K$ and stacking recovers $\hat{\boldsymbol{B}}_i$ directly. The row-independence assumption enters at $\boldsymbol{\Gamma}_i^{(k)}$ for the mean update and grabbing $(\boldsymbol{S}_i^{-1})_{kk}$ in the covariance update.

Implementation convention. The equations above are written in the observation space of $\boldsymbol{z}_i$, so $\boldsymbol{M}_\tau$ appears explicitly. The implementation instead premultiplies the observation and its noise by $\boldsymbol{M}_\tau^{-1}$, tracking $\boldsymbol{M}_\tau^{-1}\hat{\boldsymbol{z}}_i$ with noise covariance $\boldsymbol{M}_\tau^{-1}\boldsymbol{R}_i\boldsymbol{M}_\tau^{-\top}$, so no $\boldsymbol{M}_\tau$ appears anywhere in the update. The two are identical: writing $\boldsymbol{S}_i'$ for the predictive covariance in the transformed space,

$$\boldsymbol{S}_i' = \operatorname{diag}(\sigma^2_{1|\boldsymbol{u}_i}, \ldots, \sigma^2_{K|\boldsymbol{u}_i}) + \boldsymbol{M}_\tau^{-1}\boldsymbol{R}_i\boldsymbol{M}_\tau^{-\top} = \boldsymbol{M}_\tau^{-1}\boldsymbol{S}_i\boldsymbol{M}_\tau^{-\top}, \qquad \boldsymbol{S}_i'^{-1} = \boldsymbol{M}_\tau^\top \boldsymbol{S}_i^{-1}\boldsymbol{M}_\tau,$$

so $(\boldsymbol{S}_i'^{-1})_{kk} = \alpha_{i,k}$ exactly, and with $\boldsymbol{\eta}_i' = \boldsymbol{M}_\tau^{-1}\boldsymbol{\eta}_i$ the mean update $\hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i \boldsymbol{e}_k^\top \boldsymbol{S}_i'^{-1} \boldsymbol{\eta}_i'$ reduces to $\boldsymbol{g}_{i,k}\boldsymbol{\eta}_i$. Note this equivalence relies on $\boldsymbol{D}_i$ carrying the $\boldsymbol{M}_\tau \cdot \boldsymbol{M}_\tau^\top$ sandwich; pairing a flat $\boldsymbol{D}_i$ with an $\boldsymbol{\eta}_i$ that includes $\boldsymbol{M}_\tau$ gives a filter mismatched by one factor of $\boldsymbol{M}_\tau$.

Log-determinant with hypothetical $\boldsymbol{u}_{i+1}$

For a hypothetical stimulation input $\boldsymbol{u}_{i+1}$, define the one-trial-ahead predictive covariance of the observed perturbation as

$$\begin{aligned}\boldsymbol{S}_i(\boldsymbol{u}_{i+1}) &:= \mathrm{Cov}\left(\boldsymbol{z}_{i+1} \mid \mathcal{D}_i, \boldsymbol{u}_{i+1}\right) \\ &= \boldsymbol{M}_\tau \,\mathrm{diag}\left(\boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(1)} \boldsymbol{u}_{i+1}, \ldots, \boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(K)} \boldsymbol{u}_{i+1}\right) \boldsymbol{M}_\tau^\top + \boldsymbol{R}_i.\end{aligned}$$

Because

$$\boldsymbol{z}_{i+1} = \boldsymbol{M}_\tau \boldsymbol{B} \boldsymbol{u}_{i+1} + \boldsymbol{\varepsilon}_{i+1},$$

the cross-covariance between the kth parameter row and the observation vector is

$$\begin{aligned}\boldsymbol{\Gamma}_{i+1}^{(k)} &:= \mathrm{Cov}\left(\boldsymbol{w}^{(k)}, \boldsymbol{z}_{i+1}\right) \\ &= \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}_{i+1} \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \in \mathbb{R}^{N \times K}.\end{aligned}$$

The Gaussian conditioning covariance update is therefore

$$\begin{aligned}\hat{\boldsymbol{\Sigma}}_{B_{i+1}}^{(k)} &= \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} - \boldsymbol{\Gamma}_{i+1}^{(k)} \boldsymbol{S}_i(\boldsymbol{u}_{i+1})^{-1} \boldsymbol{\Gamma}_{i+1}^{(k)\top} \\ &= \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} - \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}_{i+1} \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i(\boldsymbol{u}_{i+1})^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k \boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}.\end{aligned}$$

Define

$$\begin{aligned}\alpha_{i,k}(\boldsymbol{u}_{i+1}) &:= \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i(\boldsymbol{u}_{i+1})^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k, \\ \boldsymbol{v}_k &:= \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}_{i+1}.\end{aligned}$$

Then each row's covariance update is a rank-1 downdate,

$$\hat{\boldsymbol{\Sigma}}_{B_{i+1}}^{(k)} = \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} - \alpha_{i,k}(\boldsymbol{u}_{i+1}) \boldsymbol{v}_k \boldsymbol{v}_k^\top.$$

Thus, by the matrix determinant lemma,

$$\begin{aligned}\det\left(\hat{\boldsymbol{\Sigma}}_{B_{i+1}}^{(k)}\right) &= \left[1 - \alpha_{i,k}(\boldsymbol{u}_{i+1}) \boldsymbol{v}_k^\top \left(\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\right)^{-1} \boldsymbol{v}_k\right] \det\left(\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\right) \\ &= \left[1 - \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i(\boldsymbol{u}_{i+1})^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k \, \boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}_{i+1}\right] \det\left(\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\right),\end{aligned}$$

where

$$\boldsymbol{v}_k^\top \left(\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\right)^{-1} \boldsymbol{v}_k = \boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}_{i+1}.$$

Information gain

Define the per-row information gain for a hypothetical stimulus $\boldsymbol{u}$ as

$$\begin{aligned}\mathcal{I}_{i,k}(\boldsymbol{u}) &:= \log\det\left(\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\right) - \log\det\left(\hat{\boldsymbol{\Sigma}}_{B_{i+1}}^{(k)}\right) \\ &= -\log\left[1 - \boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i(\boldsymbol{u})^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k \, \boldsymbol{u}^\top \hat{\boldsymbol{\Sigma}}_{B_i}^{(k)} \boldsymbol{u}\right].\end{aligned}$$

The relevant precision for row k is therefore

$$\boldsymbol{e}_k^\top \boldsymbol{M}_\tau^\top \boldsymbol{S}_i(\boldsymbol{u})^{-1} \boldsymbol{M}_\tau \boldsymbol{e}_k,$$

This integrates the dynamics-induced mixing through $\boldsymbol{M}_\tau$, observation uncertainty through $\boldsymbol{R}_i$, and parameter uncertainty through $\boldsymbol{D}_i$, and therefore depends on the candidate stimulus itself through $\boldsymbol{S}_i(\boldsymbol{u})$.

Although the rows of $\boldsymbol{B}$ are independent a priori, conditioning on the vector-valued observation $\boldsymbol{z}_i$ generally induces posterior correlations between rows, particularly when $\boldsymbol{M}_\tau$ mixes latent coordinates or $\boldsymbol{R}_i$ is non-diagonal. We therefore maintain only the marginal covariance of each row and discard the posterior cross-row covariances. The resulting D-optimal criterion is an approximate, row-factorized design criterion.

$$\begin{aligned}\boldsymbol{s}_{i+1}^{*} &= \arg\max_{r\in\{1,\dots,N-7\}} \sum_{k=1}^{K} -\log\left[1-\boldsymbol{e}_k^{\top}\boldsymbol{M}_\tau^{\top}\boldsymbol{S}_i(\boldsymbol{u}_r)^{-1}\boldsymbol{M}_\tau\boldsymbol{e}_k\,\boldsymbol{u}_r^{\top}\hat{\boldsymbol{\Sigma}}_{B_i}^{(k)}\boldsymbol{u}_r\right],\\ \boldsymbol{u}_r &:= \operatorname{softplus}\left(\boldsymbol{V}\boldsymbol{s}_r+\boldsymbol{b}\right),\end{aligned}$$

where

$$\boldsymbol{s}_r := \sum_{j=0}^{7}\boldsymbol{e}_{r+j}, \qquad r\in\{1,\dots,N-7\}.$$

Choosing the next stimulation pattern s_{i+1}

Both criteria reduce to maximizing a set function of the binary pattern $\boldsymbol{s} \in \{0,1\}^N$ subject to a sparsity budget, with the nonlinearity $\boldsymbol{u} = \mathrm{softplus}(\boldsymbol{V}\boldsymbol{s} + \boldsymbol{b})$ in between:

$$\boldsymbol{s}^*_{i+1} = \arg\max_{\|\boldsymbol{s}\|_0 \le 8} f(\boldsymbol{u}(\boldsymbol{s})), \qquad f_{\mathrm{D}} = \sum_k \log\big(1 + \rho_{i,k}\big), \qquad f_{\mathrm{A}} = \mathrm{tr}\big(\boldsymbol{S}^{-1}\boldsymbol{J}^\top \boldsymbol{J}\big).$$

Exhaustive search is $\binom{N}{8}$ and out of reach, so:

- **Greedy**: add the single channel with the largest marginal gain, recompute, repeat 8 times. $\log\det$ objectives of this form are submodular in the selected set, so greedy carries the standard $(1 - 1/e)$ guarantee; the trace objective does not in general.
- **Projected / proximal descent**: relax $\boldsymbol{s}$ to $[0,1]^N$ and descend, projecting to the sparse binary set either at the end (projected) or at each step (proximal).
- Both need the $\boldsymbol{V}$ spread matrix and softplus in the chain rule, since neighbouring channels are not independent knobs.

The gradient of f_{D} with respect to $\boldsymbol{u}$ is available in closed form, $\nabla_{\boldsymbol{u}} f_{\mathrm{D}} = \sum_k \frac{2\,\hat{\boldsymbol{\Sigma}}^{(k)}_{B_i}\boldsymbol{u}}{\boldsymbol{u}^\top \hat{\boldsymbol{\Sigma}}^{(k)}_{B_i}\boldsymbol{u} + r_{i,k}}$, which makes the relaxed problem cheap.

Superseded: earlier precision-form sketch

For adaptive selection, before each trial $i+1$ we want $\boldsymbol{u}_{i+1}$ maximizing the expected information gain about $\boldsymbol{B}$, i.e. minimizing the determinant of the covariance of $\boldsymbol{B}$ after trial $i+1$. Writing $\boldsymbol{\Delta}_{i+1} = \boldsymbol{B}\boldsymbol{u}_{i+1}$ for the induced latent perturbation and β for how the data is scaled relative to the prior, the precision form is

$$\boldsymbol{\Sigma}^{-1}_{B_{i+1|i}} = \boldsymbol{\Sigma}^{-1}_{B_{i|i}} + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top,$$

and the matrix determinant lemma plus $\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}] = \hat{\boldsymbol{B}}_i \boldsymbol{u}_{i+1}$ gives the objective

$$\arg\max_{\boldsymbol{u}} \; \beta\, \boldsymbol{u}^\top_{i+1} \hat{\boldsymbol{B}}_i^\top \boldsymbol{\Sigma}^B_i\, \hat{\boldsymbol{B}}_i \boldsymbol{u}_{i+1}.$$

TODO

1. We will need a **compositional model** of some sort to handle the fact that if there is behavioral change, there is likely a change in the latent dynamics as well, which will affect the estimation of $\boldsymbol{B}$. Change point detection? Or construct using known time-scale of behavioral / neural feedback
2. We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation):
3. Might need a **greedy search** for the acquisition function to not focus on the same noisy channels repeatedly.
4. This is for one timepoint, don't worry too much about some kind of t to t+k information update, before deciding on the next step, where t+k is the next stim LINOCS?
5. Do we need to expand to NKxNK for the precision matrix? instead of a K X N x N matrix.
6. Tried fitting the PLDS with a step term ($\boldsymbol{C} \times \text{step}(t)$) before, but doesn't seem to do well.
7. Try to find closed form after plugging in $\mathbb{E}[\boldsymbol{B}]$?
8. Compare A-optimal (full covariance) against D-optimal (per row) in simulation — note the two scores are not on a common scale, so compare recovery of $\boldsymbol{B}$, not objective values.
9. Use the RTS lag-one cross-covariance in $\boldsymbol{R}_i$ rather than dropping it.

- Since s is Bernoulli (discrete), projected or coordinate or proximal descent to solve for $\boldsymbol{u}$
- Projected is projecting the final to discrete, proximal is projecting each step

Random Sampling Stopgap

- Total configurations: N
- Take N samples: stimulate groups of 2 sites (one row).
- For 128 sites, this is 64 rows.
- Each stim randomly sample 20%: 12 rows (can revisit this number)
- Collect all the data and responses and see if we can predict in a leave-one-out approach (use all but one response).
- Benefit of random sampling is that we can collect now before Nike's implant comes out.